

Self-Check Questions: Miscellaneous Topics

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Contents

1	The Gamma Function	2
2	Derivative of the Inverse Function	4
3	Newton's (Newton–Raphson) Method	6
4	Euler's Method	8
5	Roots of Unity	10
6	Complex Numbers as Matrices / Representation Theory	12
7	Hidden Constants in Integration by Parts	14
8	Geometric Meaning of the Third Derivative	16
9	Second Derivative Test and Hessian	18
10	Cauchy–Schwarz Inequality	20

1 The Gamma Function

1. What is the definition of the Gamma function $\Gamma(x)$ for $x > 0$?
2. For which values of x is the integral definition of $\Gamma(x)$ valid?
3. Show that $\Gamma(1) = 1$.
4. Prove the functional equation $\Gamma(x+1) = x\Gamma(x)$.
5. How does the Gamma function extend the factorial function? State the relationship between $\Gamma(n)$ and $(n-1)!$ for $n \in \mathbb{N}$.
6. Compute $\Gamma(2)$, $\Gamma(3)$, and $\Gamma(4)$ using the functional equation.
7. What is the value of $\Gamma(\frac{1}{2})$?
8. How can the integral $\int_0^\infty e^{-t^2} dt$ be used to compute $\Gamma(\frac{1}{2})$?
9. What substitution transforms the Gaussian integral into the Gamma function integral?
10. What is the reflection formula for the Gamma function?
11. Use the reflection formula to compute $\Gamma(\frac{1}{2})$.
12. What is the duplication formula (Legendre's formula) for the Gamma function?
13. What is the behavior of $\Gamma(x)$ near $x = 0$?
14. Does the Gamma function have any poles? If so, where are they located?
15. Is the Gamma function defined for negative non-integer values? What happens at negative integers?
16. What is the logarithmic derivative of the Gamma function called?
17. Define the digamma function $\psi(x)$.
18. What is the integral representation of $\Gamma(x)$ involving $e^{-t}t^{x-1}$?
19. What is an alternative integral representation of $\Gamma(x)$ using a limit (Euler's limit form)?
20. What is Stirling's approximation for $\Gamma(x)$ for large x ?
21. How can Stirling's formula be used to approximate $n!$?
22. What is the Beta function $B(x, y)$?
23. How is the Beta function related to the Gamma function?
24. Show that $B(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$.
25. What is the integral definition of the Beta function?
26. How can you derive the Beta-Gamma relationship using a change of variables?
27. What is the asymptotic growth rate of $\Gamma(x)$ as $x \rightarrow \infty$?
28. Is the Gamma function convex or concave on $(0, \infty)$?
29. What is the logarithmic convexity (log-convexity) property of $\Gamma(x)$?
30. State Bohr-Mollerup theorem and explain its significance.
31. How does the Gamma function behave under scaling transformations?
32. What is the value of $\Gamma(\frac{3}{2})$?

33. Compute $\Gamma\left(\frac{5}{2}\right)$ using recursion.
34. Express $\Gamma\left(n + \frac{1}{2}\right)$ in terms of factorials and $\sqrt{\pi}$.
35. What is the relation between $\Gamma(x)$ and improper integrals in probability theory?
36. How does the Gamma function appear in the definition of the Gamma distribution?
37. What is the probability density function of the Gamma distribution?
38. What role does $\Gamma(x)$ play in normalizing probability distributions?
39. How can $\Gamma(x)$ be analytically continued to complex numbers?
40. Is $\Gamma(x)$ an entire function? Why or why not?
41. Where are the zeros of the Gamma function located?
42. What is the Weierstrass product representation of $\Gamma(x)$?
43. How does $\Gamma(x)$ relate to complex analysis and contour integrals?
44. What is Euler's constant γ and how does it relate to $\Gamma(x)$?
45. What is the series expansion of $\log \Gamma(x)$ near $x = 1$?
46. How is the Gamma function used in evaluating definite integrals?
47. Give an example of an integral that evaluates to a Gamma function.
48. How does the substitution $t = -\ln u$ relate the Gamma function to integrals over $(0, 1)$?
49. What is the connection between $\Gamma(x)$ and factorial moments?
50. Why is the Gamma function important in physics and engineering?

2 Derivative of the Inverse Function

1. What is the definition of the inverse of a function f ? Under what condition does an inverse exist?
2. What does it mean for a function to be one-to-one (injective), and why is this property necessary for the existence of an inverse?
3. State the relationship between a function and its inverse using composition.
4. If $y = f(x)$, how do you express x in terms of f^{-1} ?
5. What is the geometric relationship between the graphs of f and f^{-1} ?
6. How can you determine whether a function has an inverse using the horizontal line test?
7. State the formula for the derivative of an inverse function:

$$(f^{-1})'(x) = ?$$

8. Derive the formula for $(f^{-1})'(x)$ starting from the identity $f(f^{-1}(x)) = x$.
9. If $f(a) = b$, express $(f^{-1})'(b)$ in terms of $f'(a)$.
10. Under what conditions on f does the derivative of the inverse function exist?
11. Why must $f'(a) \neq 0$ for the inverse function to be differentiable at $b = f(a)$?
12. Compute $(f^{-1})'(x)$ when $f(x) = x^3 + x$.
13. Let $f(x) = \sin x$ restricted to $[-\frac{\pi}{2}, \frac{\pi}{2}]$. What is $f^{-1}(x)$ and its derivative?
14. Find the derivative of $\arctan(x)$ using the inverse function derivative formula.
15. Show that the derivative of $\ln x$ can be derived from the inverse of e^x .
16. If $f'(x)$ is known, describe the steps to compute $(f^{-1})'(x)$ at a specific point.
17. What happens to $(f^{-1})'(x)$ if $f'(x)$ is very large? What is the geometric interpretation?
18. What happens to $(f^{-1})'(x)$ if $f'(x)$ is very small (but nonzero)?
19. Explain why the slopes of inverse functions at corresponding points are reciprocals.
20. If f is decreasing and invertible, what can you say about the sign of $(f^{-1})'(x)$?
21. Let $f(x) = x^5$. Compute $(f^{-1})'(x)$ explicitly.
22. If $f(x) = \tan x$ on $(-\frac{\pi}{2}, \frac{\pi}{2})$, compute the derivative of its inverse.
23. Verify that $\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}$ using the inverse function rule.
24. How does implicit differentiation relate to finding the derivative of an inverse function?
25. If f is differentiable and strictly monotonic on an interval, what can be said about the differentiability of f^{-1} ?
26. Give an example of a function that is invertible but whose inverse is not differentiable at some point.
27. What is the domain of f^{-1} in terms of the range of f ?
28. What is the range of f^{-1} in terms of the domain of f ?
29. How can you use the inverse function derivative formula to avoid explicitly finding the inverse?

30. Let $f(2) = 5$ and $f'(2) = 3$. Find $(f^{-1})'(5)$.

3 Newton's (Newton–Raphson) Method

1. What is the goal of a root-finding algorithm?
2. State the Newton method iteration formula for finding a root of a function $f(x)$.
3. How do you obtain the next iterate x_{n+1} from the tangent line?
4. Derive the Newton iteration by applying a first-order Taylor expansion of $f(x)$ around x_n .
5. What assumptions about $f(x)$ are required for Newton's method to work?
6. What is meant by an initial guess x_0 , and why is it important?
7. Define what it means for Newton's method to converge.
8. What is quadratic convergence?
9. Why does Newton's method converge faster near the root than far from it?
10. What happens if $f'(x_n) = 0$ during the iteration?
11. Why is Newton's method considered a local method?
12. Why can Newton's method fail to converge for some initial guesses?
13. Give an example of a function and initial guess where Newton's method diverges.
14. What is meant by the basin of attraction of a root?
15. How does the choice of initial guess affect which root Newton's method converges to?
16. What is a multiple root, and how does it affect convergence of Newton's method?
17. How can Newton's method be modified to handle multiple roots?
18. What stopping criteria can be used to terminate Newton's method?
19. Compare stopping criteria based on $|x_{n+1} - x_n|$ and $|f(x_n)|$.
20. How is Newton method formulated in terms of a vector-valued function $F(x)$?
21. How can Newton's method be extended to systems of nonlinear equations?
22. How do we define the Jacobian matrix $J_F(x)$ for a function $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$?
23. Write the Newton iteration formula for solving $F(x) = 0$ in $\mathbb{R}^n$.
24. What role does the Jacobian matrix play in multidimensional Newton's method?
25. What issues arise when the Jacobian is singular or nearly singular?
26. What is a damped (or modified) Newton method?
27. Why might damping improve convergence?
28. How can Newton's method be used to find extrema of a function?
29. How is Newton's method related to solving $f'(x) = 0$?
30. What is the difference between Newton's method for optimization and for root finding?
31. What is the order of convergence of Newton's method compared to secant and bisection methods?
32. What is the secant method, and how does it approximate Newton's method?
33. When might the secant method be preferred over Newton's method?

34. How can automatic differentiation help in implementing Newton's method?
35. What is line search in the context of Newton's method?
36. How can Newton's method behave near inflection points?
37. What happens if the function is not differentiable at the root?
38. Why can Newton's method oscillate between points instead of converging?
39. How can you detect divergence in practice?
40. What are hybrid methods that combine Newton's method with bracketing methods?
41. Why are hybrid methods often preferred in practice?
42. What is the role of scaling in Newton's method for systems?
43. What is meant by the error $e_n = x_n - r$, and how does it evolve in Newton's method?
44. Derive the error recurrence relation for Newton's method near a simple root.
45. What is the goal of Newton's method in the context of optimization?
46. How do we derive Newton's method for finding a local minimum of a scalar function $f(x)$?
47. What role does the gradient $\nabla f(x)$ play in Newton's method?
48. What is the Hessian matrix $\nabla^2 f(x)$ and how is it defined?
49. Write the Newton iteration formula for optimization in $\mathbb{R}^n$.
50. How does Newton's method extend to complex-valued functions?
51. What are basins of attraction in the complex plane?
52. How can we visualize the convergence behavior of Newton's method in the complex plane?
53. How do Newton fractals arise from iterating Newton's method in $\mathbb{C}$?
54. Why do the boundaries between basins of attraction exhibit fractal structure?
55. What are fixed points of the Newton iteration map?
56. Are all fixed points of the Newton map roots of $f(z)$?
57. What are attracting, repelling, and neutral fixed points in this context?

4 Euler's Method

4.1 Fundamental Concepts

1. What problem does Euler's method aim to solve? State the general form of an initial value problem.
2. What is meant by a numerical approximation to a solution of an ODE?
3. What role does the step size h play in Euler's method?

4.2 Geometric and Intuitive Understanding

4. What is the geometric interpretation of Euler's method in terms of tangent lines?
5. How would you visualize the steps of Euler's method on a slope field?

4.3 Derivation and Formulation

6. Starting from the differential equation $y'(x) = f(x, y)$, derive the Euler update formula.
7. Write the recursive formula for Euler's method, defining all variables involved.
8. Given an initial condition $y(x_0) = y_0$, how do you compute y_1 using Euler's method?
9. How do you compute y_{n+1} from y_n in Euler's method?

4.4 Error Analysis and Accuracy

10. What is the local truncation error in Euler's method?
11. What is the global truncation error in Euler's method?
12. What is the order of accuracy of Euler's method?
13. How does decreasing the step size h affect the accuracy of Euler's method?
14. What is the trade-off between accuracy and computational cost in Euler's method?
15. What types of errors can accumulate in Euler's method over many steps?
16. How can you estimate the error in Euler's method during computation?

4.5 Stability and Limitations

17. What is meant by stability in the context of numerical methods for ODEs?
18. Why can Euler's method become unstable for certain differential equations?
19. Give an example of a differential equation where Euler's method performs poorly.
20. What is a stiff differential equation, and why is Euler's method problematic for such equations?
21. What happens if the function $f(x, y)$ is not smooth? How does this affect Euler's method?
22. What are common pitfalls when using Euler's method in practice?

4.6 Variants and Improvements

23. What is the difference between explicit Euler and implicit Euler methods?
24. Write the update formula for the implicit Euler method.
25. What advantage does the implicit Euler method have over the explicit version?
26. How can you modify Euler's method to improve accuracy (e.g., midpoint or improved Euler methods)?
27. What is the idea behind the Heun method (improved Euler method)?

4.7 Theoretical Connections

28. How does Euler's method relate to the Taylor series expansion of the solution?
29. What term(s) of the Taylor series does Euler's method include or neglect?

4.8 Extensions and Generalizations

30. Can Euler's method be used for systems of differential equations? How?
31. How does Euler's method behave when applied backward in time (negative step size)?

4.9 Euler's Method as Integration

32. What is the standard form of an ordinary differential equation involving a function $y(t)$?
33. How can the differential equation $\frac{dy}{dt} = f(t, y)$ be rewritten as an equivalent integral equation?
34. What does the integral $\int_{t_n}^{t_{n+1}} f(s, y(s)) ds$ represent conceptually?
35. Why is the integral form of the solution exact, even though we may not know $y(s)$?
36. What is the main difficulty in computing the integral $\int_{t_n}^{t_{n+1}} f(s, y(s)) ds$ exactly?
37. What approximation is made in the Forward Euler method for the integrand $f(s, y(s))$?
38. What is the geometric interpretation of approximating $f(s, y(s))$ by $f(t_n, y(t_n))$?
39. How does this approximation simplify the integral over the interval $[t_n, t_{n+1}]$?
40. How do we define the step size h , and what role does it play?
41. How do we derive the Forward Euler update formula from the approximated integral?
42. Why can we interpret the Forward Euler method as a left-endpoint Riemann sum?
43. In what sense is Forward Euler performing "integration" of the differential equation?
44. Why does assuming a constant slope over the interval introduce error?

5 Roots of Unity

1. What is meant by an n th root of unity?
2. Write the equation whose solutions are the n th roots of unity.
3. How many distinct n th roots of unity are there? Why?
4. Express the n th roots of unity in exponential form using Euler's formula.
5. Write the general formula for the k th n th root of unity.
6. What is the principal (or primitive) n th root of unity?
7. How can all n th roots of unity be generated from a single primitive root?
8. What is the difference between a root of unity and a primitive root of unity?
9. State the condition under which a root of unity is primitive.
10. How many primitive n th roots of unity are there? Which function counts them?
11. What is Euler's totient function $\varphi(n)$ and how does it relate to roots of unity?
12. Represent the n th roots of unity geometrically in the complex plane.
13. What is the magnitude and argument of each n th root of unity?
14. How are the n th roots of unity distributed on the complex unit circle?
15. What is the angle between consecutive n th roots of unity?
16. Compute the sum of all n th roots of unity. What is the result?
17. Why does the sum of all n th roots of unity equal zero?
18. Compute the product of all n th roots of unity.
19. What symmetry properties do the n th roots of unity exhibit?
20. Show that the n th roots of unity form a group under multiplication.
21. What type of group is formed by the n th roots of unity?
22. What is the identity element in this group?
23. What is the inverse of a given n th root of unity?
24. Show that the set of n th roots of unity is cyclic.
25. What is a generator of the group of n th roots of unity?
26. For which values of k is ζ^k a primitive root, where ζ is a primitive n th root?
27. Solve the equation $z^n = 1$ explicitly for small values such as $n = 2, 3, 4$.
28. What are the cube roots of unity? Write them explicitly.
29. What are the fourth roots of unity? Write them explicitly.
30. How can roots of unity be used to factor the polynomial $x^n - 1$?
31. Write the factorization of $x^n - 1$ in terms of its complex roots.
32. What are cyclotomic polynomials and how are they related to roots of unity?
33. Define the n th cyclotomic polynomial $\Phi_n(x)$.
34. How does $x^n - 1$ factor into cyclotomic polynomials?

35. What is the relationship between primitive roots of unity and cyclotomic polynomials?
36. How can roots of unity be used to evaluate sums such as $\sum_{k=0}^{n-1} \zeta^k$?
37. Evaluate $\sum_{k=0}^{n-1} \zeta^{mk}$ for an integer m .
38. Under what condition does $\sum_{k=0}^{n-1} \zeta^{mk} = 0$?
39. What happens when m is a multiple of n in the above sum?
40. How are roots of unity used in discrete Fourier transforms (DFT)?
41. What is the connection between roots of unity and periodicity?
42. How can roots of unity simplify computations involving periodic sums?
43. Show that if ζ is an n th root of unity, then $\bar{\zeta} = \zeta^{-1}$.
44. What is the complex conjugate of a root of unity geometrically?
45. How do roots of unity relate to rotations in the complex plane?
46. What is the minimal polynomial of a primitive n th root of unity?
47. How does the degree of $\Phi_n(x)$ relate to $\varphi(n)$?
48. In what fields do primitive n th roots of unity lie?
49. What is the significance of roots of unity in number theory?
50. How are roots of unity used in solving polynomial equations?

6 Complex Numbers as Matrices / Representation Theory

<https://www.youtube.com/watch?v=hsveVFoIJPM>

1. What does it mean to represent an abstract algebraic object (like a group element or number) as a matrix?
2. What is the general idea of a **representation** in the context of linear algebra and group theory?
3. How can a complex number $a + bi$ be represented as a 2×2 real matrix?
4. Write explicitly the matrix corresponding to the complex number $a + bi$.
5. Show how matrix addition corresponds to addition of complex numbers under this representation.
6. Show how matrix multiplication corresponds to multiplication of complex numbers under this representation.
7. Why is this matrix representation of complex numbers structure-preserving?
8. What algebraic structure is preserved when mapping complex numbers to matrices?
9. What property must a mapping satisfy to be considered a homomorphism?
10. Verify that the mapping from complex numbers to matrices is a homomorphism.
11. What is the identity element in the complex numbers, and what matrix represents it?
12. What matrix corresponds to the imaginary unit i ?
13. Compute the square of the matrix representing i and interpret the result.
14. How does this matrix representation encode the fact that $i^2 = -1$?
15. What geometric transformation in $\mathbb{R}^2$ corresponds to multiplication by i ?
16. How does multiplication by a general complex number $a + bi$ act geometrically on the plane?
17. How is rotation represented in terms of matrices?
18. How does scaling appear in the matrix representation of a complex number?
19. Explain how complex multiplication combines rotation and scaling.
20. What is the determinant of the matrix corresponding to $a + bi$?
21. What is the geometric meaning of this determinant?
22. What is the trace of the matrix representation of $a + bi$?
23. How does the modulus $|a + bi|$ relate to the determinant of the matrix?
24. What condition on a and b makes the matrix invertible?
25. What is the matrix corresponding to the inverse of a complex number?
26. Show that the inverse matrix corresponds to the multiplicative inverse of the complex number.
27. How does this representation help visualize complex numbers as linear transformations?
28. What is the connection between this representation and rotations in the plane?

29. How can this idea be generalized to represent other algebraic structures using matrices?
30. What is a group representation in general?
31. Why are matrices particularly useful for representing abstract algebraic objects?
32. How does this example motivate the study of representation theory?
33. What advantages do matrix representations provide for computation?
34. How can eigenvalues of these matrices be interpreted in terms of complex numbers?
35. What happens when you diagonalize the matrix corresponding to a complex number?
36. How does this representation connect linear algebra with complex analysis?
37. In what sense are complex numbers “the same” as a subset of 2×2 real matrices?
38. What is the dimension of the vector space in which these matrices act?
39. How does this perspective change your understanding of multiplication by complex numbers?
40. Can every linear transformation of $\mathbb{R}^2$ be represented by a complex number? Why or why not?
41. What distinguishes matrices that correspond to complex numbers from arbitrary 2×2 matrices?
42. How would you test whether a given 2×2 matrix corresponds to a complex number?
43. What deeper insight does this example give about the relationship between algebra and geometry?

7 Hidden Constants in Integration by Parts

<https://www.youtube.com/watch?v=N5xtRr0jfWI>

<https://www.youtube.com/watch?v=V8uZYdd4Jnk>

1. Why does every indefinite integral include an arbitrary constant C ?
2. Derive the integration by parts formula from the product rule.
3. Why is the equation

$$\int u \, dv = uv - \int v \, du$$

technically incomplete if we ignore constants of integration? Where are the hidden constants in this equation?

4. Why can ignoring arbitrary constants sometimes lead to paradoxes or false identities when manipulating indefinite integrals?
5. Consider this example

$$I = \int \underbrace{\frac{f'(x)}{f(x)}}_{uv'} dx = \underbrace{\frac{f(x)}{f(x)}}_{uv} - \int \underbrace{-\frac{f'(x)}{f^2(x)}}_{u'} \underbrace{f(x)}_v dx = 1 + \int \frac{f'(x)}{f(x)} dx = 1 + I$$

$$\begin{aligned} u &= \frac{1}{f(x)} & v' &= f'(x) \\ u' &= -\frac{f'(x)}{f(x)^2} & v &= f(x) \end{aligned}$$

Where is the error in this paradoxical derivation?

6. Consider this example

$$I = \int f(x)g'(x) \, dx$$

$$\begin{aligned} u &= f(x) & v' &= g'(x) \\ u' &= f'(x) & v &= g(x) + A \end{aligned}$$

Show how all the constants of integration are absorbed into one when applying integration by parts.

7. Why can repeated applications of integration by parts accumulate multiple constants?
8. How can these multiple constants always be absorbed into a single constant in the final answer?
9. Why is it sometimes useful to deliberately choose a non-zero constant when computing an antiderivative inside integration by parts?
10. If the integration by part is applied naively to integral $\int x \arctan(x) \, dx$ it leads to an incomplete solution

$$\int \underbrace{x}_{v'} \underbrace{\arctan(x)}_u dx = \underbrace{\frac{x^2}{2}}_v \underbrace{\arctan(x)}_u - \int \underbrace{\frac{x^2}{2}}_v \underbrace{\frac{1}{x^2+1}}_{u'} dx.$$

How can choosing a convenient constant in the antiderivative simplify the naive solution?

11. If the integration by part is applied naively to integral $\int \ln(x+2) dx$ it leads to an incomplete solution

$$\int \underbrace{1}_{v'} \underbrace{\ln(x+2)}_u dx = \underbrace{x}_v \underbrace{\ln(x+2)}_u - \int \underbrace{x}_v \underbrace{\frac{1}{x+2}}_{u'} dx.$$

How can choosing a convenient constant in the antiderivative simplify the naive solution?

12. Why are definite integrals immune to the constant-of-integration ambiguity?

8 Geometric Meaning of the Third Derivative

<https://www.youtube.com/watch?v=SovllrJUQ64>

1. What is the geometric meaning of the first derivative $f'(x)$ in terms of the graph of $f(x)$?
2. How can you interpret the second derivative $f''(x)$ geometrically?
3. What does it mean for $f''(x) > 0$ in terms of the shape of the graph?
4. What does it mean for $f''(x) < 0$?
5. Define an inflection point in terms of concavity.
6. What condition on $f''(x)$ is necessary for an inflection point?
7. Why is $f''(x) = 0$ not sufficient to guarantee an inflection point?
8. What additional condition ensures that a point where $f''(x) = 0$ is actually an inflection point?
9. How can sign changes of $f''(x)$ be used to detect inflection points?
10. What is the geometric meaning of the third derivative $f'''(x)$?
11. How can $f'''(x)$ be interpreted as a rate of change of another geometric quantity?
12. If $f'''(x) > 0$, what does this say about how concavity is changing?
13. If $f'''(x) < 0$, what does this imply about the change in concavity?
14. How does $f'''(x)$ describe the “bending behavior” of a curve beyond concavity?
15. Explain how the third derivative relates to the steepness of the tangent slope.
16. How does the graph of $f'(x)$ help visualize $f''(x)$?
17. How does the graph of $f''(x)$ help visualize $f'''(x)$?
18. If $f''(x)$ has a local maximum or minimum, what can you say about $f'''(x)$ at that point?
19. What is the relationship between $f'''(x) = 0$ and extrema of $f''(x)$?
20. Can $f'''(x) = 0$ correspond to an inflection point of $f(x)$? Explain.
21. How would you distinguish between:
 - an inflection point of $f(x)$
 - an extremum of $f''(x)$
22. Describe how successive derivatives correspond to successive “rates of change” geometrically.
23. Give a physical interpretation of $f'(x)$, $f''(x)$, and $f'''(x)$ in motion (position, velocity, acceleration, ...).
24. What is the physical meaning of the third derivative in kinematics (often called jerk)?
25. How does the sign of the third derivative affect motion in a physical system?
26. Construct an example function where $f''(x) = 0$ but there is no inflection point.
27. Construct an example where $f'''(x) = 0$ but $f''(x)$ does not change sign.
28. Given a graph, how would you visually estimate where $f'''(x)$ is positive or negative?
29. How can you use higher derivatives to understand increasingly subtle features of a curve?

30. Why is the third derivative rarely discussed compared to the first and second in basic calculus?
31. In what types of problems does understanding the third derivative become important?

9 Second Derivative Test and Hessian

<https://www.youtube.com/watch?v=z0bSdV8DA4M>

9.1 Core Concept: What is the second derivative test really doing?

- What does the second derivative test attempt to determine about a function at a critical point?
- Why is checking only the first derivative insufficient for classifying critical points?
- In one variable, what does the sign of $f''(x)$ tell us geometrically?
- How can you interpret $f''(x)$ in terms of curvature?

9.2 From 1D to Multivariable Functions

- What is the analogue of the second derivative for multivariable functions?
- What is the Hessian matrix, and how is it constructed?
- Why can't we just look at "the second derivative" in multiple dimensions?
- What does the Hessian represent geometrically?

9.3 Quadratic Approximation

- What is the second-order Taylor expansion of a function near a critical point?
- Why does the linear term vanish at a critical point?
- What role does the quadratic term play in determining local behavior?
- How does the function locally resemble a quadratic form?

9.4 Quadratic Forms and Geometry

- What is a quadratic form?
- How can the expression $\mathbf{x}^T H \mathbf{x}$ describe local curvature?
- What does it mean for a quadratic form to be:
 - Positive definite?
 - Negative definite?
 - Indefinite?

9.5 Classification of Critical Points

- How does the definiteness of the Hessian classify a critical point?
- What condition corresponds to a local minimum?
- What condition corresponds to a local maximum?
- What condition corresponds to a saddle point?
- What happens if the Hessian is degenerate (determinant zero)?

9.6 Eigenvalues Interpretation

- Why are eigenvalues the key to understanding the Hessian?
- How do eigenvalues relate to curvature along different directions?
- What does it mean if all eigenvalues are positive?
- What does it mean if eigenvalues have mixed signs?
- How do eigenvectors relate to principal directions of curvature?

9.7 Geometric Intuition

- Why is a saddle point associated with “curving up in some directions and down in others”?
- How can you visualize curvature along different directions through a point?
- Why is the second derivative test really a statement about curvature in all directions?

9.8 Connection to Linear Algebra

- Why can any symmetric matrix be diagonalized?
- How does diagonalizing the Hessian simplify the quadratic form?
- What does the diagonal form reveal about the function locally?

9.9 Deeper Understanding

- Why is the second derivative test fundamentally about approximating the function by a quadratic surface?
- In what sense is the test coordinate-independent?
- How does this perspective generalize beyond $\mathbb{R}^2$ or $\mathbb{R}^3$?

9.10 Edge Cases and Limitations

- Why does the test fail when eigenvalues are zero?
- What additional analysis is needed in degenerate cases?
- Can higher-order terms change the classification?

10 Cauchy–Schwarz Inequality

10.1 Basic Definitions and Geometric Meaning

- State the Cauchy–Schwarz inequality for vectors in $\mathbb{R}^n$.
- State the inequality for arbitrary inner product spaces.
- What conditions must a function $\langle \cdot, \cdot \rangle$ satisfy to be an inner product?
- Explain geometrically what the inequality says about two vectors.
- Why does the inequality imply that

$$|\cos \theta| \leq 1$$

for the angle between vectors?

- Derive the formula

$$\cos \theta = \frac{\langle u, v \rangle}{\|u\| \|v\|}$$

from the inequality.

- Compute both sides of the inequality for

$$u = (1, 2), \quad v = (3, -4).$$

- Verify the inequality for

$$u = (1, 1, 1), \quad v = (2, 0, -1).$$

- What happens if one of the vectors is the zero vector?
- Why is the inequality trivial in that case?

10.2 Equality Conditions

- Under what conditions does equality hold in the Cauchy–Schwarz inequality?
- Prove that equality holds if $u = \lambda v$ for some scalar λ .
- Prove the converse: if equality holds, then the vectors are linearly dependent.
- Give examples of:
 1. vectors for which equality holds,
 2. vectors for which the inequality is strict.
- Explain geometrically why parallel vectors produce equality.
- Can orthogonal vectors produce equality? Why or why not?

10.3 Proof Techniques

- Prove the inequality using the nonnegativity of

$$\|u - tv\|^2.$$

- In that proof, why must the discriminant of the resulting quadratic polynomial be nonpositive?

- Expand

$$\|u - tv\|^2$$

explicitly and identify the quadratic coefficients.

- Give a proof using orthogonal projection.
- Give a proof using minimization.
- Why does positivity of norms play a central role in every proof?
- Which assumptions about the inner product are actually used in the standard proof?

10.4 Complex Inner Product Spaces

- State the Cauchy–Schwarz inequality for complex vector spaces.
- Why must absolute values appear around the inner product?
- What changes in the proof for complex spaces?
- Verify the inequality for

$$u = (1 + i, 2), \quad v = (i, 1 - i).$$

- Why is

$$\langle u, u \rangle$$

always real and nonnegative even in complex spaces?

- Explain why

$$\langle u, v \rangle \neq \langle v, u \rangle$$

in general for complex inner products.

10.5 Connections to Other Inequalities

- Derive the triangle inequality from Cauchy–Schwarz.
- Show how Cauchy–Schwarz implies

$$(a_1 + \cdots + a_n)^2 \leq n(a_1^2 + \cdots + a_n^2).$$

- Derive the inequality

$$\left(\sum_{k=1}^n a_k b_k \right)^2 \leq \left(\sum_{k=1}^n a_k^2 \right) \left(\sum_{k=1}^n b_k^2 \right).$$

10.6 Function Spaces and Integrals

- State the integral version of the Cauchy–Schwarz inequality.
- Verify the inequality for

$$f(x) = x, \quad g(x) = 1 - x$$

on $[0, 1]$.

- Show that

$$\left(\int_a^b f(x)g(x) dx \right)^2 \leq \left(\int_a^b f(x)^2 dx \right) \left(\int_a^b g(x)^2 dx \right).$$

- What inner product naturally appears in function spaces?
- What does equality mean in the integral version?
- Why is the integral version important in Fourier analysis?

10.7 Problem-Solving and Exploration

- Find vectors $u, v \in \mathbb{R}^2$ such that

$$\langle u, v \rangle = 0.$$

- Find nonzero vectors satisfying equality in Cauchy–Schwarz.
- Construct an example where the inequality is very far from equality.
- Use Cauchy–Schwarz to prove

$$(a + b + c)^2 \leq 3(a^2 + b^2 + c^2).$$

- Use the inequality to bound

$$|x + 2y + 3z|.$$

- Find the maximum possible value of

$$x + 2y$$

subject to

$$x^2 + y^2 = 1.$$

- Show that the angle definition from inner products is well-defined.
- Explain why projections minimize distance.
- Derive the least-squares approximation formula using orthogonality and Cauchy–Schwarz.
- Investigate whether analogous inequalities exist for norms that are not induced by inner products.